

Tensor-based Opposing yet Complementary Learning for Multi-view Multi-label Feature Selection

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Abstract

Multi-view multi-label learning (MVML) is a significant area of research in multimedia, providing a foundational framework for various real-world applications. However, the richness of its descriptive capabilities often results in high-dimensional data that contains redundant information, negatively affecting model performance. Most existing methods do not thoroughly explore the mapping of the distinctive parts while balancing common and distinctive information. Additionally, there has been limited focus on the relationships between different types of mappings and the high-order constraints among view-specific labels. In this paper, we propose a novel tensor-based method for view-specific label learning that integrates adaptive weight mechanisms into both global non-linear and local linear mappings. This method effectively captures high-order relationships among views and hybrid labels through hierarchical label correlation constraints. Central to our model is the “Opposing yet Complementary” procedure, which enhances feature weight representation at a finer-grained level. Extensive experiments on widely used multi-view multi-label datasets demonstrate significant performance improvements, underscoring the effectiveness of our proposed method.

CCS Concepts

• **Computing methodologies** → **Machine learning methods**; • **Information systems** → *Information systems applications*.

Keywords

Multi-view learning, Multi-label learning, Tensor nuclear norm, Feature selection

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1 Introduction

Multi-view multi-label learning (MVML) deals with scenarios where each data instance is described by multiple heterogeneous feature sets and is linked to several class labels simultaneously [2, 37, 41]. Due to the inherent characteristic of MVML, which allows for comprehensive representation of instances by capturing its rich information, it has been widely applied in various fields such as bioinformatics [1] and video analysis [36]. However, this complexity also introduces the challenge of managing high-dimensional data. As a result, feature selection becomes a crucial pre-processing step aimed at identifying the most relevant features, thereby improving classifier performance and scalability. This technique has proven particularly effective in areas like image annotation [32], person re-identification [55, 66], action recognition [61] and security [38, 48], where intricate relationships are prevalent with high-dimension.

As the key point of dealing with the problem in MVML, effectively leveraging the rich information from multiple views remains a significant challenge. Initially, some methods focus on identifying the commonality among different views [33, 57]. An alternative method emphasizes the use of label information to bolster the discriminative power of the learned feature subspace [35]. Despite these advancements in MVML, the common information remains insufficient to comprehensively represent all views [39]. Thus, integrating both distinctive and common information offers a promising type of methods to exploring comprehensive data representations. From the perspective of label types, one line of research focuses on the observed labels. In this type, it is assumed that the standard label for each view is consistent, meaning that the label for each view is aligned with a shared reference label [28].

Based on this assumption, to enhance the extraction of distinctive parts, one effective way is to explicitly construct and learn the individuality and commonality [26]. For instance, Tan et al. [46] emphasizes exploring the underlying geometric structure of the features, while Lyu et al. [40] concentrates on enforcing constraints that capture the consensus and exclusive semantic information across views. Another way to improve the performance is to concatenate each view as a complete one to preserve global information [67]. However, the directly concatenated ignore the relationship between views. Thus, some research like [15, 16] emphasizes the different importance of each view, and introduces distinctive information from different perspectives.

Contrasting with the above studies, the second line uses the observed label as the label for all views, meaning the label for each view is not fully identified [62, 68]. Thus, this imprecise mapping across views often degrades feature selection performance.

An alternative type learns view-specific labels derived from the observed label but explicitly formulates them within the objective function [16], enabling precise mapping between features and their corresponding view-specific labels. By removing the constraint of aligning each view, this method allows for more accurate determination of feature weights for individual views [64]. However, some researches such as [63] utilize a two-stage procedure that may experience performance degradation if the supervisory information derived from the first stage is inaccurate. Recent research has shifted toward united models that better capture inter-view relationships [14, 29]. Despite competitive results, prior studies often overlook different types of the mapping and higher-order constraints among view-specific labels, which could enhance improvements.

To overcome the limitations of existing methods, we propose a novel tensor-based method for view-specific label learning in multi-view multi-label feature selection in Fig. 1, termed TOCL. The primary objective of TOCL is to establish accurate mappings between features and labels within each view, ensuring the retention of informative features essential for downstream tasks. Thus, the precision of view-specific labels is emphasized, aiming to infer them accurately from the observed labels. Our method captures high-order relationships through tensor constraints, effectively retaining the principal components of multi-view data. The construction of view-specific labels is inherently guided by the observed labels to ensure consistency. Additionally, during the learning phase, the local learning process focuses on extracting distinctive features, while the global learning process mitigates feature redundancy across views by assessing feature weights across all views. This dual process allows for a finer-grained understanding of view importance to model diverse relationships. Although these processes may appear contradictory, they are mutually reinforcing, which is why we characterize our method as “Opposing yet Complementary”. Extensive experiments conducted on six real-world multi-view datasets validate the effectiveness of TOCL, demonstrating superior performance compared to several state-of-the-art methods. The main contribution of this paper is summarized as follows.

- This paper establishes the constant and high-order relationships related to view-specific labels and the non-linear mappings from feature space to labels, resulting in enhanced label learning performance and further refinement of the associated feature weight learning.
- We integrate global and local perspectives for feature-label mapping and introduce an instance importance matrix. This matrix guides the learning process across different views, emphasizing the distinctions between views within the overall space.
- An alternating optimization algorithm with multiplicative update rules to solve the model both effectively and efficiently. Extensive experiments on six multi-view datasets demonstrate the superiority of our method.

2 Related Work

2.1 Tensor-based Multi-view Method

Tensor-based multi-view methods mainly utilize tensor representations. The Tensor Nuclear Norm (TNN) enhances low-rank tensor

recovery and captures high-order interactions [21, 45], thus improving model robustness against noise and missing data [6, 42]. The foundational model for TNN based on t-SVD is as follows:

$$\begin{aligned} \min_{Z^{(v)}, E^{(v)}} \quad & \lambda \|E\|_{2,1} + \|Z\|_*, \\ \text{s.t.} \quad & X^{(v)} = X^{(v)} Z^{(v)} + E^{(v)}, \quad v = 1, 2, \dots, V, \\ & Z = \phi(Z^{(1)}, Z^{(2)}, \dots, Z^{(V)}), \\ & E = [E^{(1)}, E^{(2)}, \dots, E^{(V)}], \end{aligned} \quad (1)$$

where the input data $X^{(v)} \in \mathbb{R}^{d_v \times N}$, and $Z^{(v)} (v = 1, 2, \dots, V)$ are transformed into Z which is a 3-mode tensor through the function $\phi(\cdot)$. The function $[\cdot]$ is designed to enforce the consistency of columns across each view. The $l_{2,1}$ -norm and the TNN based on t-SVD are denoted as $\|\cdot\|_{2,1}$ and $\|\cdot\|_*$.

Recent advances in tensor-based methods have explored multiple directions to enhance model performance. One key line of work focuses on designing objective functions tailored to specific problem structures, enabling more effective optimization [3, 4, 27]. Another emphasizes leveraging the 3-D nature of tensors, where slices from different angles capture distinct semantics for richer representation [43, 43, 44]. Tensor norms have also been extensively studied, with variants like the Schatten- p norm offering adaptability through tunable parameters [5, 11, 23, 51]. In addition, some works enhance local geometry and inter-view correlations via structural regularization [56], while others tackle data noise with robustness-oriented designs [50]. Motivated by these methods, our work primarily focuses on extracting the principal components of the tensor by eliminating view redundancy and sample noise.

2.2 Multi-view Multi-label learning

Current MVML methods primarily aim to balance consistency and complementarity for comprehensive data representation [54, 58]. Some works, such as Lyu et al. [41], enhance robustness by jointly modeling feature and view neighborhoods in a unified self-representation. However, methods like [52] often neglect the challenge of high-dimensional feature selection, limiting their applicability.

In real-world scenarios, challenges such as incomplete views, partial labels, non-aligned settings, and non-overlapping scenarios are frequently encountered [22, 49, 65]. To address these issues, various methods have been proposed [25, 65]. For instance, Li et al. [25] align different views by identifying shared labels. Jointly tackling incomplete views and partial labels, Liu et al. [30, 31] propose two distinct methods by employing a cross-channel contrastive learning and graph attention network separately. In contrast, Xie et al. [53] adopt an uncertainty-aware pseudo-labeling strategy to impute missing labels, followed by dual graph constraints imposed on features using the imputed labels. Additionally, Tan et al. [47] leverage information theory within a deep neural network framework to balance cross-view mutual information and view-specific information. Despite these advancements, there remains a lack of in-depth exploration into the mapping of distinctive parts, which could further enhance the understanding and handling of these complex scenarios.

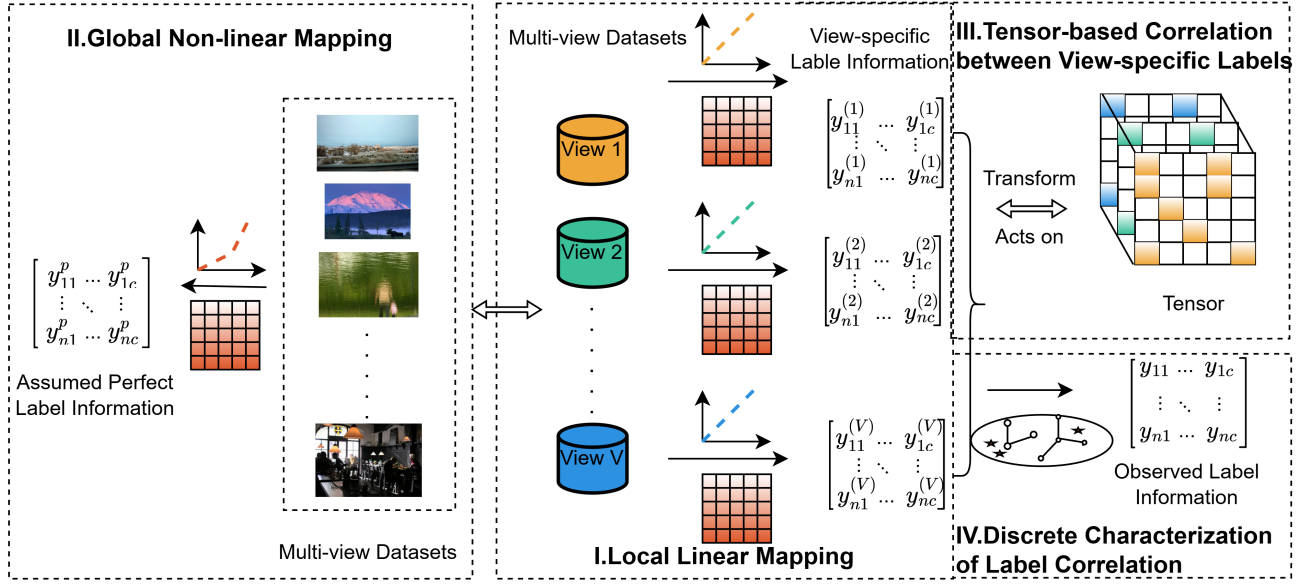


Figure 1: An overview of the proposed TOCL in the field of multi-view multi-label learning is presented. Our task is to effectively order and select relevant features in the high-dimension space. We aim to uncover latent relationships within multi-view data and enhance label learning performance. Specifically, TOCL is composed of four key components: 1) Local Linear Mapping with Instance-level Importance; 2) Global Non-linear Mapping; 3) Tensor-based Correlation between View-specific Labels; 4) Discrete Characterization of Label Correlation.

3 Notations

Throughout this research, we employ various notations to describe the problems in MVML. Considering the property for the supervised learning and multiple views, the input data can be divided into two parts. One is the view space as $X^{(i)} \in \mathbb{R}^{n \times d^{(i)}}$ where $i = 1, 2, \dots, V$, and V denotes the number of views. The other is the label space as $Y \in \{0, 1\}^{n \times l}$, which is observed from realistic scene. The lowercase letters for n, l and d represent the number of instances, labels and features. The supervised learning is defined between the two kinds of space, which means that $x_{(m.)}^{(i)} \in X^{(i)}$ associated with the j -th label sets $y_{(m.)}$ as 1.

Definition 1 (Tensor Singular Value Decomposition (t-SVD)): We assume that the two tensors $\mathcal{U} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ and $\mathcal{V} \in \mathbb{R}^{n_2 \times n_2 \times n_3}$ are satisfied with the orthogonal request. Meanwhile, an f-diagonal tensor with diagonal matrices on its frontal slices is represented as $\mathcal{S} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$. Then, the t-SVD of $\mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ can be expressed as:

$$\mathcal{A} = \mathcal{U} * \mathcal{S} * \mathcal{V}^T, \quad (2)$$

where the conjugate transpose of a tensor $\mathcal{V} \in \mathbb{R}^{n_1 \times n_3 \times n_2}$ is the tensor $\mathcal{V}^T \in \mathbb{R}^{n_2 \times n_1 \times n_3}$.

Definition 2 (Tensor Nuclear Norm based on t-SVD): The TNN of the tensor is calculated as the sum of the nuclear norms of all its frontal slices after applying the Discrete Fourier transform (DFT) along the third dimension:

$$\|\mathcal{A}\|_* = \sum_{i=1}^r S(i, i, 1) = \sum_{i=1}^r \sum_{j=1}^{n_3} \bar{S}(i, i, j), \quad (3)$$

where $\|\cdot\|_*$ denotes the nuclear norm, and r is the tensor tubal rank of $\mathcal{A}$. The result of DFT on $\mathcal{A}$ along the 3-rd dimension can be represented as $\bar{\mathcal{S}} \in \mathbb{C}^{n_1 \times n_2 \times n_3}$.

Definition 3 (First-layer label correlation): Given a multi-view multi-label data set, $Y_{all} \in \{0, 1\}^{n \times l}$ denotes the merged matrix of view-specific labels, while $Y_n \in \{0, 1\}^{n \times l}$ denotes noisy labels for all views. The relationship between these matrices and the observed label matrix Y is defined as follows:

$$Y = Y_{all} \otimes Y_n, \quad (4)$$

where the operation ' $\otimes$ ' represents that an element occupying the identical positional index is marked.

Definition 4 (Second-layer label correlation): The label matrix specific to each view is characterized as $y_t^{(i)} \in \{0, 1\}^{n \times l}$.

$$Y_{all} = y_t^{(1)} \otimes y_t^{(2)} \dots \otimes y_t^{(V)}. \quad (5)$$

4 The Proposed Method

Unlike existing methods, TOCL focuses on establishing an accurate mapping between the feature space and the view-label space. This is achieved through constraints that enforce local linear mapping with instance-level importance. Additionally, it incorporates global non-linear mapping and tensor-based correlations between view-specific labels, while capturing the discrete characterization of label correlations. Subsequently, we employ multiplicative update rules to address the optimization problem and analyze the complexity of the proposed method.

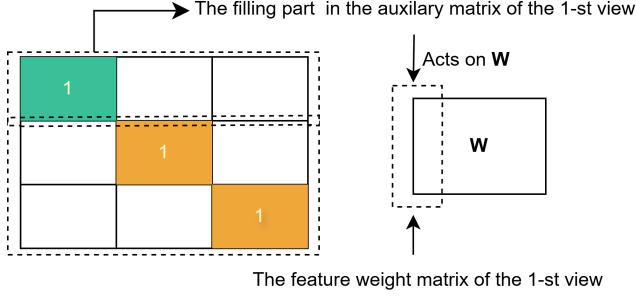


Figure 2: The transformation process from the combined feature weights of all views to the individual feature weights for each view.

4.1 Local Linear Mapping with Instance-level Importance

The feature ordering is determined by constructing the feature weight, which allows for the computation of feature importance [18]. In contrast to traditional loss functions in multi-label learning, the feature space comprises multiple subspaces, and each corresponding to a distinct view with view-specific labels replacing observed labels. Whereas the goal from the perspective of feature selection is to choose features across all views, feature ordering is computed by integrating information from each view. Previous studies [46] suggest aggregating feature weights from different views, weighted by their respective importance. However, this type of method requires learning additional hyperparameters for view importance, increasing the complexity of the learning process.

To address this, we revert to constructing the model with $W \in \mathbb{R}^{d \times l}$, the transformation process is designed to capture the relationship between the variables $U^{(i)}$ and W . As illustrated in Fig. 2, the number of views is set to two, with the corresponding feature weight matrices $U^{(0)}$ and $U^{(1)}$ containing one and two features, respectively. The auxiliary matrices $A^{(0)}$ and $A^{(1)}$ are then designed to generate the feature weight matrices for each view, where the diagonal elements are set to '1' based on the number of features from the previous views. The transformation is defined as follows.

$$U^{(i)} = A^{(i)}W. \quad (6)$$

On this basis, to ensure a thorough exploration of the distinctive aspects while fully utilizing various information, the importance matrix P is designed to adapt to each view. This matrix allows for nuanced data analysis by preserving the unique characteristics of each view and enhancing performance through the capture of intricate patterns and relationships within the data. Consequently, the regression error between the feature space and the label space is minimized for each view. The instance-level importance is incorporated as follows.

$$\mathcal{L}_1(\cdot) = \sum_{i=1}^V \left\| PX^{(i)}A^{(i)}W - y_t^{(i)} \right\|_F^2. \quad (7)$$

4.2 Global Non-linear Mapping

To gain a more complex understanding of the relationship between features and labels, we incorporate the sigmoid function into the regularization process. This enables a non-linear mapping from features to labels, transforming the output of a linear function into a probability. Therefore, the nonlinear relationship between features and the perfect label matrix Y_p can be expressed as:

$$Y_p = \frac{I}{I + e^{-XV}}U, \quad (8)$$

where $V \in \mathbb{R}^{d \times k}$ is the projection coefficient and $U \in \mathbb{R}^{k \times l}$ is the association matrix related to latent labels. Although the instance-level importance matrix P is to adjust the weight for each views, a limitation exists in that the correlation among lviews is not considered. In other words, during the local mapping process, P exhibits increased sensitivity in capturing the differences among individual views. In contrast, during the global mapping process, P prioritizes global optimization while ensuring the stability of the iterative learning process. Thus, considering both the global perspective and the nonlinear mapping, the $\mathcal{P}_1(\cdot)$ can be reformulated as:

$$\mathcal{P}_1(\cdot) = \left\| PXW - \frac{I}{I + e^{-XV}}U \right\|_F^2 + \|U\|_F^2 + \|V\|_F^2 + \|P\|_F^2 + \|W\|_{2,1}. \quad (9)$$

4.3 Tensor-based Correlation between View-specific Labels

In multi-view learning, it is assumed that the common information across all views constitutes most in each individual view. Thus, the concept of co-occurrence patterns [10] is introduced, while also considering the constraint imposed by intra-view label correlation. Meanwhile, based on this assumption, the third-order tensor is expected to be of low rank. Additionally, leveraging the inherent properties of low rank also allows for denoising matrices.

Unfortunately, even so, a limitation remains in that the fixed threshold τ in this method fails to recognize that larger singular values, which capture the principal directions of the data. These values should be penalized less to preserve critical information. Thus, the variable weights $w_i^{(j)}$ for different singular values is introduced into the tensor nuclear norm. This enhances the flexibility of the norm and enables a more refined treatment of the singular values. The formulation can then be summarized as follows.

$$\mathcal{L}_2(\cdot) = \|\mathcal{Z} - \mathcal{T}\|_F^2 + \|\mathcal{T}\|_{w,*}, \quad (10)$$

let $\mathcal{T} \in \mathbb{R}^{n \times V \times n} = \text{transpose}\left(\left\{y_t^{(i)}, y_t^{(i)T}\right\}_{i=1}^V, (1 : 3 : 2)\right)$ where the dimension of $\mathcal{Z}$ matches that of $\mathcal{T}$ to help capture the primary components of $\mathcal{T}$. Meanwhile, the tensor singular value thresholding (t-SVT) operator $D_\tau(\cdot)$ [10] is applied to $\mathcal{Z}$ to maintain the low-rank property of $\mathcal{T}$, and the process is defined as:

$$D_\tau(\mathcal{Z}) = \mathcal{U} * \mathcal{S}_\tau * \mathcal{V}^T. \quad (11)$$

4.4 Discrete Characterization of Label Correlation

From the perspective of all view labels, a key challenge is the accurate identification and mitigation of noise in the observed label

matrix. Although constraints on $y_t^{(i)}$ are set when exploring high-order relationships and interactions with latent labels, the constant relationships are overlooked, which impacts the learning direction. First, we introduce a novel metric ' $\ominus$ ', to quantify the element-wise divergence between two matrices. Specifically, the divergence between the observed label matrix Y and the merged matrix of view-specific labels Y_{all} is defined as:

$$y_{(pq)} \ominus y_{all(pq)} = \begin{cases} 1 & y_{(pq)} \neq y_{all(pq)} \\ 0 & y_{(pq)} = y_{all(pq)} \end{cases}. \quad (12)$$

This metric is applied element-wise across the matrices, providing a binary indicator of whether the observed and inferred labels differ at each position. To improve the resilience of the model against noise, we introduce an F -norm regularization term. This term encourages sparsity in the noise matrix Y_n , effectively reducing the impact of noisy labels and facilitating their suppression. The penalty function is formulated as:

$$\mathcal{P}_2(\cdot) = \|Y_{all} \otimes Y_n \ominus Y\|_F^2 + \|Y_n\|_F^2. \quad (13)$$

4.5 The Objective Function

By jointly integrating $\mathcal{L}_1(\cdot)$, $\mathcal{L}_2(\cdot)$, $\mathcal{P}_1(\cdot)$ and $\mathcal{P}_2(\cdot)$ into a unified framework, the objective function is formulated as follows.

$$\begin{aligned} \min_{y_t^{(i)}, P, W, U, V, Y_n, \mathcal{T}} & \underbrace{\sum_{i=1}^V \left\| PX^{(i)} A^{(i)} W - y_t^{(i)} \right\|_F^2 + \|\mathcal{Z} - \mathcal{T}\|_F^2 + \tau \|\mathcal{T}\|_*}_{\text{instance-level mapping with tensor constraint}} \\ & + \alpha \left(\left\| PXW - \frac{I}{I + e^{-XV}} U \right\|_F^2 + \|U\|_F^2 + \|V\|_F^2 \right), \\ & + \underbrace{\beta \|P\|_F^2 + \gamma \left(\|Y_{all} \otimes Y_n \ominus Y\|_F^2 + \|Y_n\|_F^2 \right) + \delta \|W\|_{2,1}}_{\text{enhanced function for splitting labels and mapping}}, \end{aligned} \quad (14)$$

where α, β, γ and δ are trade-off parameters that balance the model. From the objective function in Formula (14), it can be seen that P is mainly governed by two terms: one responsible for the local linear mapping and the other for the global non-linear mapping. The role of P is to bridge the gap between these two mappings. Regarding $y_t^{(i)}$, the specific label matrix, it is subject to three constraints, which include mapping to the feature space and enforcing hierarchical correlation among labels. These constraints ensure that an accurate label space is learned for each view. As for Y_n , the noise in the label space is addressed through regularization in the first-layer label correlation. Thus, through the accurate learning of each variable, the feature weight W is refined, leading to better results that contribute to more accurate ranking.

4.6 Optimization

To solve the problem in Formula (14), an optimization algorithm based on multiplicative update rules is designed. In the solution procedure, we impose constraints on the non-negativity of variables. In each iteration, all variables except the one being solved for are fixed, due to the non-convex of the problem. This allows the problem to be decomposed into several subproblems. The pseudocode is presented in Algorithm 1.

Algorithm 1 Tensor-based Opposing yet Complementary Learning

Input: Data matrices $\{X^{(i)}\}_{i=1}^V$, and Label matrix Y .

Parameter: Parameters α, β, γ and δ .

Output: Selected features.

- 1: Initialize $W, P, Y_n, U, V, y_t^{(i)}, \mathcal{T}$;
- 2: **repeat**
- 3: Update the matrix $W, P, Y_n, U, V, y_t^{(i)}$ according to Formula (16)-(21);
- 4: Update the matrix $\mathcal{T}$ according to Formula (25);
- 5: Update the objective function (14);
- 6: **until** Convergence
- 7: Obtain the ordered feature sequence by calculating $\|W_j\|_2$ where $j = 1, 2, 3, \dots, d$;
- 8: **return** Top ranked features as $s\text{-TOCL-f}$.

Step1-W: Given the smoothness of the problem for the $l_{2,1}$ -norm, the transformation for related terms such as $\|W\|_{2,1} = 2\text{Tr}(W^T D W)$ can be applied. The corresponding part of W in the objective function is then described as follows:

$$\begin{aligned} \min_W \sum_{i=1}^V & \left\| PX^{(i)} A^{(i)} W - y_t^{(i)} \right\|_F^2 + \alpha \left\| PXW - \frac{I}{I + e^{-XV}} U \right\|_F^2 \\ & + 2\delta \text{Tr}(W^T D W) - \text{Tr}(\varphi W^T), \end{aligned} \quad (15)$$

where $\varphi \in \mathbb{R}^{d \times c}$ is a multiplier, and we set $F = I/(I + e^{-XV})$. The derivative is set to zero. Then, the variable W can be updated by:

$$W \leftarrow W \circ \frac{\sum_{i=1}^V A^{(i)T} X^{(i)T} P^T y_t^{(i)} + \alpha X^T P^T F U}{\sum_{i=1}^V A^{(i)T} X^{(i)T} P^T P X^{(i)} A^{(i)} W + \alpha X^T P^T P X W + 2\delta D W}. \quad (16)$$

Similarly, the variable $P, Y_n, U, V, y_t^{(i)}$ can be updated as follows.

$$P \leftarrow P \circ \frac{\sum_{i=1}^V y_t^{(i)} W^T A^{(i)T} X^{(i)T} + \alpha F U W^T X^T}{\sum_{i=1}^V P X^{(i)} A^{(i)} W W^T A^{(i)T} X^{(i)T} + \alpha B W W^T X^T + \beta P}, \quad (17)$$

$$Y_n \leftarrow Y_n \circ \frac{Y \ominus \sum_{i=1}^V y_t^{(i)}}{2Y_n}, \quad (18)$$

$$U \leftarrow U \circ \frac{F^T P X W}{F^T F U + U}, \quad (19)$$

$$V \leftarrow V \circ \frac{X^T E P X W U^T}{X^T E F U U^T + V}, \quad (20)$$

$$y_t^{(i)} \leftarrow y_t^{(i)} \circ \frac{P X^{(i)} A^{(i)} W + T^{(i)} y_t^{(i)} + \gamma C}{(1 + \gamma) y_t^{(i)} + y_t^{(i)} y_t^{(i)T} y_t^{(i)}}, \quad (21)$$

where $B = P X$, $E = e^{-XV}/(I + e^{-XV})^2$ and $C = \sum_{i=1}^V y_t^{(i)} - y_t^{(i)}$.

Step2- $\mathcal{T}$: To simplify the optimization, the problem is transformed into the Fourier domain, and the Formula (10) is changed as follows.

$$\min_{\tilde{\mathcal{T}}(:, :, i)} \frac{1}{n} \sum_{j=1}^n \|\tilde{\mathcal{Z}}(:, :, j) - \tilde{\mathcal{T}}(:, :, j)\|_F^2 + \|\tilde{\mathcal{T}}(:, :, j)\|_{w,*}. \quad (22)$$

Table 1: The detailed information for the datasets in our experiments.

Views	SCENE	OBJECT	MIRFlickr	Corel5K	IAPRTC12	3Sources
View-1(d_1)	CH(64)	CH(64)	DH(100)	DH(100)	DH(100)	BBC(1000)
View-2(d_2)	CM(225)	CM(225)	GIST(512)	DHV3H1(300)	DHV3H1(300)	Reuters(1000)
View-3(d_3)	CORR(144)	CORR(144)	HH(100)	GIST(512)	GIST(512)	Guardian(1000)
View-4(d_4)	EDH(73)	EDH(73)	-	HHV3H1(300)	HHV3H1(300)	-
View-5(d_5)	WT(128)	WT(128)	-	HH(100)	HH(100)	-
Instances(n)	4400	6047	4053	4999	4999	169
Features(d)	634	634	712	1312	1312	3000
Labels(l)	33	31	38	260	260	6

For each slice $\tilde{\mathcal{Z}}(:, :, j)$, the SVD is computed as:

$$\tilde{\mathcal{Z}}(:, :, j) = \tilde{\mathcal{U}}(:, :, j) \tilde{\mathcal{S}}(:, :, j) \tilde{\mathcal{V}}(:, :, j)^T. \quad (23)$$

The singular value $\tilde{\mathcal{S}}(:, :, j)$ is computed as:

$$\tilde{\mathcal{S}}(:, :, j) = \begin{cases} 0 & c_2 < 0 \\ \frac{c_1 + \sqrt{c_2}}{2} & c_2 \geq 0 \end{cases}, \quad (24)$$

where $c_1 = \tilde{\mathcal{S}}(i, i, j) - \epsilon$ and $c_2 = (\tilde{\mathcal{S}}(i, i, j) - \epsilon)^2 - 4C$, and ϵ is a small positive value for numerical stability. C is a regularization parameter, and the weight is $w_i^{(j)} = C / (\tilde{\mathcal{S}}(i, i, j) + \epsilon)$. The updated slice $\tilde{\mathcal{T}}(:, :, j)$ is then reconstructed using the singular values:

$$\tilde{\mathcal{T}}(:, :, j) = \tilde{\mathcal{U}}(:, :, j) \tilde{\mathcal{S}}(:, :, j) \tilde{\mathcal{V}}(:, :, j)^T. \quad (25)$$

4.7 Complexity Analysis

The optimization procedure involves seven key components, with the dimension of the feature matrix for each view assumed to be d . The computational complexity for updating the variables primarily involves matrix multiplications, except for $\mathcal{T}$, which also requires Fast Fourier Transform (FFT), inverse FFT (iFFT), and Singular Value Decomposition (SVD). In detail:

- For the updates of W , V , $y_t^{(i)}$, and P , the complexity is $O(n^2d)$.
- For updating $\mathcal{T}$, U , Y_n , the complexities are $O(n^2V \log n) + O(n^2V^2)$, $O(n^2k)$ and $O(nc)$ respectively.

Therefore, the overall training complexity for each iteration can be approximated as: $O(n^2d) + O(n^2V \log n) + O(n^2V^2)$ assuming $n \gg d \gg c$ and $d \gg k$.

5 Experiments

5.1 Experimental Setup

5.1.1 Datasets. We evaluate the proposed method on six commonly used multi-view multi-label datasets in Table 1, including SCENE [7], OBJECT [7], MIRFlickr [20], Corel5K [8], IAPRTC12 [9], and 3Sources [13]. The feature dimensions of these datasets range from hundreds to thousands. Missing views are denoted by the symbol ‘-’ within the table.

5.1.2 Comparing Methods. To benchmark the proposed method, we select eight representative algorithms for comparison: four multi-view multi-label methods (MSFS [62], M2LD [35], DHLI [17], EF²FS [15]) and four multi-label learning methods (MoRE [34], MRDM [19], MDFS [59], CLML [24]). These baselines are designed from distinct perspectives, such as feature-label mapping and label-relationship modeling. Despite their different design focuses, they

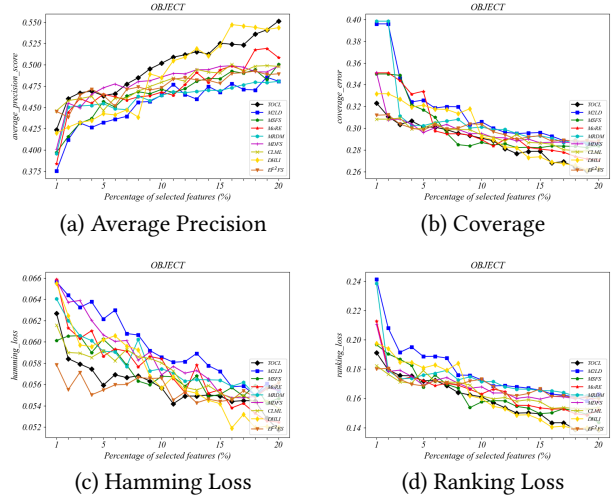


Figure 3: Nine methods on OBJECT in terms of Average Precision, Coverage, Hamming Loss and Ranking loss.

all emphasize embedded learning within the feature space to facilitate sorting tasks.

5.1.3 Evaluation Metrics. We present the evaluation results as mean accuracy along with standard deviation obtained from five-fold cross-validation. The parameters for each method are tuned within the range of $\{10^{-3}, 10^{-2}, \dots, 10^3\}$. All methods adhere to the above standards to ensure the validity of the comparative results presented in this paper, with four widely adopted metrics employed for assessment [12, 60]: Average Precision (AP), Coverage, Hamming Loss (HL), and Ranking Loss (RL). Specifically, higher AP values indicate better performance, while lower values of Coverage, HL, and RL signify superior outcomes.

5.2 Compared Results

In Tables 2 and 3, we report experiment results across all datasets under feature percentages ranging from 1% to 20%. Best results are highlighted in boldface, while second-best results are underlined. The symbol ‘-’ indicates unavailable results due to program errors. To visualize the performance of TOCL, we showcase results for one dataset across the four metrics in Fig. 3. Additionally, Table 4 presents Friedman test results to analyze method significance. These results yield the following key observations.

Table 2: Experimental results (mean $\pm$ std) in terms of AP and Coverage, where the 1st/2nd best results are shown in bold-face/underline.

Dataset	TOCL	M2LD	MSFS	MoRE	MRDM	MDFS	CLML	DHLI	EF ² FS
<i>AP $\uparrow$</i>									
SCENE	0.7981$\pm$0.015	0.7792 $\pm$ 0.014	0.7887 $\pm$ 0.008	0.7877 $\pm$ 0.007	0.7929 $\pm$ 0.01	0.7870 $\pm$ 0.007	0.7870 $\pm$ 0.008	<u>0.7951$\pm$0.012</u>	0.7935 $\pm$ 0.007
OBJECT	0.4959$\pm$0.035	0.4531 $\pm$ 0.026	0.4626 $\pm$ 0.029	0.4719 $\pm$ 0.031	0.4595 $\pm$ 0.02	0.4756 $\pm$ 0.026	0.4748 $\pm$ 0.024	<u>0.4845$\pm$0.046</u>	0.4722 $\pm$ 0.019
MIRFlickr	0.6908$\pm$0.011	0.6436 $\pm$ 0.013	0.6417 $\pm$ 0.010	0.6322 $\pm$ 0.011	0.6407 $\pm$ 0.008	<u>0.6848$\pm$0.011</u>	0.6442 $\pm$ 0.007	0.6635 $\pm$ 0.015	0.6759 $\pm$ 0.012
Corel5K	0.2419$\pm$0.008	0.2355 $\pm$ 0.008	0.1861 $\pm$ 0.007	0.1845 $\pm$ 0.023	0.2339 $\pm$ 0.011	0.2277 $\pm$ 0.014	0.2330 $\pm$ 0.008	<u>0.2370$\pm$0.009</u>	0.2282 $\pm$ 0.010
IAPRTC12	0.1510$\pm$0.008	0.1370 $\pm$ 0.006	0.1385 $\pm$ 0.005	0.1263 $\pm$ 0.006	0.1400 $\pm$ 0.005	0.1377 $\pm$ 0.006	0.1352 $\pm$ 0.002	0.1421 $\pm$ 0.007	<u>0.1509$\pm$0.010</u>
3Sources	0.4883$\pm$0.045	0.4368 $\pm$ 0.038	0.4252 $\pm$ 0.035	-1 $\pm$ 0	0.4196 $\pm$ 0.041	0.4481 $\pm$ 0.043	0.4232 $\pm$ 0.046	0.4534 $\pm$ 0.034	<u>0.4608$\pm$0.044</u>
<i>Coverage $\downarrow$</i>									
SCENE	<u>0.4193$\pm$0.018</u>	0.4416 $\pm$ 0.016	0.4271 $\pm$ 0.012	0.4295 $\pm$ 0.010	0.4252 $\pm$ 0.017	0.4331 $\pm$ 0.009	0.4308 $\pm$ 0.012	0.4174$\pm$0.017	0.4244 $\pm$ 0.014
OBJECT	0.2893$\pm$0.020	0.3142 $\pm$ 0.03	0.3005 $\pm$ 0.027	0.3020 $\pm$ 0.030	0.3090 $\pm$ 0.031	0.2993 $\pm$ 0.018	0.2980 $\pm$ 0.020	<u>0.2979$\pm$0.026</u>	0.2988 $\pm$ 0.012
MIRFlickr	0.5715$\pm$0.009	0.6069 $\pm$ 0.009	0.6173 $\pm$ 0.013	0.6397 $\pm$ 0.010	0.6128 $\pm$ 0.017	<u>0.5744$\pm$0.008</u>	0.6191 $\pm$ 0.015	0.5897 $\pm$ 0.013	0.5774 $\pm$ 0.008
Corel5K	0.4710$\pm$0.010	0.5080 $\pm$ 0.008	0.5085 $\pm$ 0.012	0.5378 $\pm$ 0.012	0.4918 $\pm$ 0.021	0.4950 $\pm$ 0.016	0.4951 $\pm$ 0.004	<u>0.4832$\pm$0.011</u>	0.4881 $\pm$ 0.019
IAPRTC12	<u>0.5072$\pm$0.041</u>	0.5131 $\pm$ 0.011	0.5098 $\pm$ 0.009	0.5877 $\pm$ 0.020	0.5097 $\pm$ 0.017	0.5187 $\pm$ 0.021	0.5149 $\pm$ 0.005	0.5091 $\pm$ 0.009	0.4794$\pm$0.016
3Sources	0.6017$\pm$0.038	0.6397 $\pm$ 0.030	0.6602 $\pm$ 0.037	-1 $\pm$ 0	0.6723 $\pm$ 0.042	0.6431 $\pm$ 0.044	0.6727 $\pm$ 0.046	<u>0.6181$\pm$0.029</u>	0.6251 $\pm$ 0.035

(1) The method TOCL on each metric has a bright performance, which ranks first in 79.17% cases. Meanwhile, the method DHLI and EF²FS also excel, ranking first or second in 58.33% and 25% of cases, respectively. This advantage stems from the consideration of global-local feature-label mappings from TOCL, which enhances mapping accuracy compared to methods like MSFS and M2LD. Unlike EF²FS, TOCL and DHLI incorporate view-specific label relationships, with TOCL further leveraging tensor nuclear norm to model high-order interactions, validating its structural design.

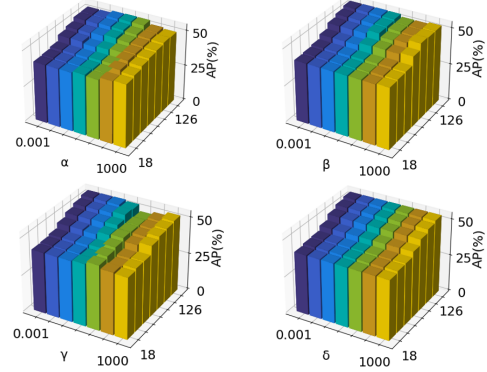
(2) Single-view algorithms underperform multi-view counterparts due to ignoring inter-view correlations. Among multi-label baselines, MDFS stands out by reconstructing the label matrix with local similarity structures, emphasizing the procedure for the label matrix purification. The non-linear description of the method MRDM outperforms the low-rank projection of the method MoRE in most cases, highlighting the advantage of non-linear relationships in modeling.

(3) As shown in Fig. 3, the AP curve rises with increasing features before stabilizing, while Coverage and RL curves decline initially and plateau. This pattern validates the effectiveness of feature selection strategy for TOCL, demonstrating its ability to leverage feature relevance for improved ranking and loss minimization.

(4) Table 4 lists critical values at $\alpha = 0.05$ and Friedman statistic F_F for each metric. The Friedman test assumes that all methods have the same performance. If the value of the metric F_F exceeds the critical value 2.244, it means that the significant differences exist between TOCL and other methods. From this table, we could observe that all metrics surpass this threshold, reinforcing its superiority.

5.3 Parameter Analysis

The TOCL framework incorporates four parameters α , β , γ and δ into two key components, including different types of mappings and hierarchical label correlations. We tune each parameter individually

**Figure 4: Parameter sensitivity studies on the OBJECT dataset.**

while keeping the others constant. Fig. 4 shows the tuning range on the X-axis, spanning from 0.001 to 1000, with the number of features depicted on the Y-axis. On the OBJECT dataset, the plot of AP metric illustrates how these four parameters impact the method. Overall, the proposed method shows minimal sensitivity to parameter changes. It exhibits only slight fluctuations in trends across different configurations which indicates the robust stability.

5.4 Convergence

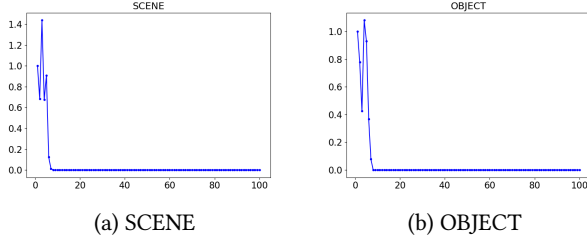
In Fig. 5, the Y-axis represents the value of $|z^{t-1} - z^t| / z^{t-1}$, where z^t indicates the objective function values at iteration t . The X-axis shows the number of iterations. The stopping criterion is met when this value becomes sufficiently small. The Y-axis setting also reflects the convergence speed. To aid in observing oscillations, the initial point is set to approximately 1. The curve rapidly decreases

Table 3: Experimental results (mean $\pm$ std) in terms of HL and RL, where the 1st/2nd best results are shown in boldface/underline.

Dataset	TOCL	M2LD	MSFS	MoRE	MRDM	MDFS	CLML	DHLI	EF ² FS
<i>HL $\downarrow$</i>									
SCENE	0.09688$\pm$0.01	0.10396 $\pm$ 0.01	0.09988 $\pm$ 0.01	0.10073 $\pm$ 0.00	0.09787 $\pm$ 0.01	0.10008 $\pm$ 0.01	0.10105 $\pm$ 0.01	<u>0.09691$\pm$0.01</u>	0.09777 $\pm$ 0.00
OBJECT	0.05600$\pm$0.00	0.05974 $\pm$ 0.00	0.05737 $\pm$ 0.00	0.05757 $\pm$ 0.00	0.05826 $\pm$ 0.00	0.05858 $\pm$ 0.00	0.05735 $\pm$ 0.00	0.05722 $\pm$ 0.00	<u>0.05601$\pm$0.00</u>
MIRFlickr	0.17056$\pm$0.01	0.20288 $\pm$ 0.01	0.19438 $\pm$ 0.00	0.19311 $\pm$ 0.00	0.19462 $\pm$ 0.00	<u>0.17191$\pm$0.01</u>	0.19173 $\pm$ 0.00	0.18084 $\pm$ 0.01	0.17438 $\pm$ 0.01
Corel5K	<u>0.01379$\pm$0.00</u>	0.01465 $\pm$ 0.00	0.01381 $\pm$ 0.00	0.01381 $\pm$ 0.00	0.01380 $\pm$ 0.00	0.01385 $\pm$ 0.00	0.01382 $\pm$ 0.00	<u>0.01379$\pm$0.00</u>	0.01374$\pm$0.00
IAPRTC12	<u>0.01543$\pm$0.00</u>	0.01653 $\pm$ 0.00	0.01561 $\pm$ 0.00	0.01550 $\pm$ 0.00	0.01560 $\pm$ 0.00	0.01537$\pm$0.00	0.01554 $\pm$ 0.00	0.01549 $\pm$ 0.00	0.01549 $\pm$ 0.00
3Sources	0.23114$\pm$0.01	0.24908 $\pm$ 0.03	0.23566 $\pm$ 0.02	-1 $\pm$ 0	0.23584 $\pm$ 0.02	0.23827 $\pm$ 0.02	0.24715 $\pm$ 0.02	<u>0.23152$\pm$0.02</u>	0.23320 $\pm$ 0.03
<i>RL $\downarrow$</i>									
SCENE	0.0914$\pm$0.01	0.1037 $\pm$ 0.009	0.0970 $\pm$ 0.006	0.0974 $\pm$ 0.005	0.0949 $\pm$ 0.007	0.0970 $\pm$ 0.006	0.0980 $\pm$ 0.006	<u>0.0926$\pm$0.008</u>	0.0947 $\pm$ 0.007
OBJECT	0.1607$\pm$0.017	0.1791 $\pm$ 0.019	0.1648 $\pm$ 0.017	0.1661 $\pm$ 0.016	0.1744 $\pm$ 0.017	0.1706 $\pm$ 0.015	0.1658 $\pm$ 0.013	<u>0.1642$\pm$0.021</u>	0.1691 $\pm$ 0.010
MIRFlickr	0.1537$\pm$0.007	0.1672 $\pm$ 0.008	0.1855 $\pm$ 0.006	0.1925 $\pm$ 0.006	0.1858 $\pm$ 0.007	<u>0.1541$\pm$0.007</u>	0.1844 $\pm$ 0.006	0.1671 $\pm$ 0.009	0.1591 $\pm$ 0.007
Corel5K	0.2198$\pm$0.006	0.2379 $\pm$ 0.005	0.2665 $\pm$ 0.009	0.2613 $\pm$ 0.008	0.2343 $\pm$ 0.015	0.2376 $\pm$ 0.012	0.2357 $\pm$ 0.003	<u>0.2283$\pm$0.007</u>	0.2347 $\pm$ 0.015
IAPRTC12	<u>0.1927$\pm$0.023</u>	0.2070 $\pm$ 0.009	0.2093 $\pm$ 0.006	0.2452 $\pm$ 0.010	0.2069 $\pm$ 0.013	0.2094 $\pm$ 0.011	0.2134 $\pm$ 0.003	0.2068 $\pm$ 0.011	0.1922$\pm$0.010
3Sources	0.4488$\pm$0.045	0.5045 $\pm$ 0.039	0.5253 $\pm$ 0.043	-1 $\pm$ 0	0.536 $\pm$ 0.051	0.4943 $\pm$ 0.047	0.5345 $\pm$ 0.055	<u>0.4761$\pm$0.035</u>	0.4862 $\pm$ 0.042

Table 4: The critical value at the significance level $\alpha = 0.05$ and Friedman statistic F_F on four evaluation metrics.

Evaluation metrics	F_F	Critical value ($\alpha = 0.05$)
Average precision	6.629	2.244
Coverage	9.100	
Hamming loss	9.730	
Ranking loss	6.248	

**Figure 5: Convergence curves analysis of TOCL on (a) SCENE and (b) OBJECT.**

in the early iterations with minimal fluctuations and then stabilizes into a consistent convergence pattern. Notably, the objective function values reach a stable state quickly, demonstrating efficient convergence behavior.

5.5 Ablation Study

To evaluate the effectiveness of the proposed method, we decompose the benchmark TOCL into four variant models. GN emphasizes regularization for the global-nonlinear mapping, while DLC maintains the constraints for discrete label correlation. In the loss function, TC retains only the tensor-based correlations among view-specific labels, and LL focuses on the local linear mapping between features and view-specific labels. Table 5 shows that the benchmark

Table 5: Ablation experimental results for AP of TOCL on SCENE, OBJECT and 3Source.

Model	Components				Datasets		
	GN	DLC	TC	LL	SCENE	OBJECT	3Sources
M_1	-	✓	✓	✓	0.7792	0.5054	0.4443
M_2	✓	-	✓	✓	0.7850	0.4750	0.3968
M_3	✓	✓	-	✓	0.7850	0.4689	0.4306
M_4	✓	✓	✓	-	0.7949	0.4841	0.4271
TOCL	✓	✓	✓	✓	0.7959	0.4959	0.4880

TOCL generally outperforms the variant algorithms. However, there is potential for improvement through the identification of more universal mapping functions. Nevertheless, these findings still indicate that removing any component negatively impacts the results of Average Precision, highlighting the critical importance of all parts in the proposed method.

6 Conclusion

In this paper, the proposed TOCL establishes opposing and complementary mappings between features and labels with instance-level importance. It captures high-order relationships among view-specific labels through the tensor constraints. Furthermore, to improve the partitioning of the label space across views, the method incorporates functions that construct relationships between different types of the label matrix such as the noisy labels, the observed labels and the view-specific labels. Comprehensive experiments and analysis prove the superiority and validity of our method. As such, this research provides valuable insights that we hope will inspire further investigation in exploring more universal fusion mapping functions for the MVML task.

Acknowledgments

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References

- [1] Robert Ebo Armah-Sekum, Sandor Szedmak, and Juho Rousu. 2024. Protein function prediction through multi-view multi-label latent tensor reconstruction. *BMC bioinformatics* 25, 1 (2024), 174.
- [2] S Bickel and T Scheffer. 2004. Multiview clustering. In *International Conference on Data Mining*, Vol. 4. 19–26.
- [3] Bing Cai, Gui-Fu Lu, Hua Li, and Weihong Song. 2024. Tensorized scaled simplex representation for multi-view clustering. *IEEE Transactions on Multimedia* 26 (2024), 6621–6631.
- [4] Zhiwen Cao and Xijiong Xie. 2024. Partition-Level Tensor Learning-Based Multi-view Unsupervised Feature Selection. *IEEE Transactions on Neural Networks and Learning Systems* (2024).
- [5] Wei Chang, Huimin Chen, Feiping Nie, Rong Wang, and Xuelong Li. 2024. Tensorized and compressed multi-view subspace clustering via structured constraint. *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2024).
- [6] Hangjun Che, Xinyu Pu, Deqiang Ouyang, and Beibei Li. 2024. Enhanced Tensorial Self-representation Subspace Learning for Incomplete Multi-view Clustering. In *Proceedings of the 32nd ACM International Conference on Multimedia*. 719–728.
- [7] Tat-Seng Chua, Jinhui Tang, Richang Hong, Haojie Li, Zhiping Luo, and Yantao Zheng. 2009. Nus-wide: a real-world web image database from national university of singapore. In *Proceedings of the ACM international conference on image and video retrieval*. 1–9.
- [8] Pinar Duygulu, Kobus Barnard, Joao FG de Freitas, and David A Forsyth. 2002. Object recognition as machine translation: Learning a lexicon for a fixed image vocabulary. In *Computer vision—ECCV 2002: 7th European conference on computer vision copenhagen, Denmark, May 28–31, 2002 proceedings, part IV* 7. Springer, 97–112.
- [9] Hugo Jair Escalante, Carlos A Hernández, Jesus A Gonzalez, Aurelio López-López, Manuel Montes, Eduardo F Morales, L Enrique Sucar, Luis Villaseñor, and Michael Grubinger. 2010. The segmented and annotated IAPR TC-12 benchmark. *Computer vision and image understanding* 114, 4 (2010), 419–428.
- [10] Linan Feng and Bir Bhanu. 2015. Semantic concept co-occurrence patterns for image annotation and retrieval. *IEEE transactions on pattern analysis and machine intelligence* 38, 4 (2015), 785–799.
- [11] Wei Feng, Zhenwei Wu, Qianqian Wang, Bo Dong, and Quanxue Gao. 2024. Federated Fuzzy C-means with Schatten-p Norm Minimization. In *Proceedings of the 32nd ACM International Conference on Multimedia*. 9407–9416.
- [12] Eva Gibaja and Sebastián Ventura. 2015. A tutorial on multilabel learning. *ACM Computing Surveys (CSUR)* 47, 3 (2015), 1–38.
- [13] Derek Greene and Pádraig Cunningham. 2009. A matrix factorization approach for integrating multiple data views. In *Joint European conference on machine learning and knowledge discovery in databases*. Springer, 423–438.
- [14] Pingting Hao, Weiping Ding, Wanfu Gao, and Jialong He. 2024. Exploring view-specific label relationships for multi-view multi-label feature selection. *Information Sciences* 681 (2024), 121215.
- [15] Pingting Hao, Wanfu Gao, and Liang Hu. 2025. Embedded feature fusion for multi-view multi-label feature selection. *Pattern Recognition* 157 (2025), 110888.
- [16] Pingting Hao, Kunpeng Liu, and Wanfu Gao. 2024. Anchor-guided global view reconstruction for multi-view multi-label feature selection. *Information Sciences* 679 (2024), 121124.
- [17] Pingting Hao, Kunpeng Liu, and Wanfu Gao. 2024. Double-layer hybrid-label identification feature selection for multi-view multi-label learning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 38. 12295–12303.
- [18] Chenping Hou, Feiping Nie, Dongyun Yi, and Yi Wu. 2011. Feature selection via joint embedding learning and sparse regression. In *IJCAI proceedings-international joint conference on artificial intelligence*, Vol. 22. Citeseer, 1324.
- [19] Rui Huang and Zhejun Wu. 2021. Multi-label feature selection via manifold regularization and dependence maximization. *Pattern Recognition* 120 (2021), 108149.
- [20] Mark J Huiskes and Michael S Lew. 2008. The mir flickr retrieval evaluation. In *Proceedings of the 1st ACM international conference on Multimedia information retrieval*. 39–43.
- [21] Jintian Ji and Songhe Feng. 2023. High-order complementarity induced fast multi-view clustering with enhanced tensor rank minimization. In *Proceedings of the 31st ACM International Conference on Multimedia*. 328–336.
- [22] Bingbing Jiang, Xingyu Wu, Xiren Zhou, Yi Liu, Anthony G Cohn, Weiguo Sheng, and Huanhuan Chen. 2022. Semi-supervised multiview feature selection with adaptive graph learning. *IEEE Transactions on Neural Networks and Learning Systems* 35, 3 (2022), 3615–3629.
- [23] Jing Li, Quanxue Gao, Qianqian Wang, Ming Yang, and Wei Xia. 2023. Orthogonal non-negative tensor factorization based multi-view clustering. *Advances in neural information processing systems* 36 (2023), 18186–18202.
- [24] Junlong Li, Peipei Li, Xuegang Hu, and Kui Yu. 2022. Learning common and label-specific features for multi-label classification with correlation information. *Pattern recognition* 121 (2022), 108259.
- [25] X Li and S Chen. 2021. A concise yet effective model for non-aligned incomplete multi-view and missing multi-label learning. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 44, 10 (2021), 5918–5932.
- [26] Y Li, J Zhang, H Wu, et al. 2025. Consistent and specific multi-view multi-label learning with correlation information. *Information Sciences* 687 (2025), 121395.
- [27] Cheng Liang, Daoyuan Wang, Huaxiang Zhang, Shichao Zhang, and Fei Guo. 2024. Robust tensor subspace learning for incomplete multi-view clustering. *IEEE Transactions on Knowledge and Data Engineering* (2024).
- [28] Bo Liu, Weinbin Li, Yanshan Xiao, Xiaodong Chen, Laiwang Liu, Changdong Liu, Kai Wang, and Peng Sun. 2023. Multi-view multi-label learning with high-order label correlation. *Information Sciences* 624 (2023), 165–184.
- [29] Bing-Qing Liu, Bin-Bin Jia, and Min-Ling Zhang. 2023. Towards enabling binary decomposition for partial multi-label learning. *IEEE transactions on pattern analysis and machine intelligence* 45, 11 (2023), 13203–13217.
- [30] C Liu, J Jia, J Wen, et al. 2024. Attention-Induced Embedding Imputation for Incomplete Multi-View Partial Multi-Label Classification. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 38. 13864–13872.
- [31] Chengliang Liu, Jie Wen, Yabo Liu, Chao Huang, Zhihao Wu, Xiaoling Luo, and Yong Xu. 2023. Masked two-channel decoupling framework for incomplete multi-view weak multi-label learning. *Advances in Neural Information Processing Systems* 36 (2023), 32387–32400.
- [32] K Liu, T Li, X Yang, et al. 2023. SemiFREE: semisupervised feature selection with fuzzy relevance and redundancy. *IEEE Transactions on Fuzzy Systems* 31, 10 (2023), 3384–3396.
- [33] Meng Liu, Yong Luo, Dacheng Tao, Chao Xu, and Yonggang Wen. 2015. Low-rank multi-view learning in matrix completion for multi-label image classification. In *Proceedings of the AAAI conference on artificial intelligence*, Vol. 29.
- [34] Siyu Liu, Xuehua Song, Zhongchen Ma, Ernest Domanaanmwi Ganaa, and Xi-angJun Shen. 2022. MoRE: multi-output residual embedding for multi-label classification. *Pattern Recognition* 126 (2022), 108584.
- [35] W Liu, J Yuan, G Lyu, and S Feng. 2023. Label driven latent subspace learning for multi-view multi-label classification. *Appl. Intell.* 53, 4 (2023), 3850–3863.
- [36] X Liu, P Zhang, C Yu, et al. 2024. A video is worth three views: Trigeminal transformers for video-based person re-identification. *IEEE Transactions on Intelligent Transportation Systems* (2024).
- [37] Yabo Liu, Jinghua Wang, Linhui Xiao, Chengliang Liu, Zhihao Wu, and Yong Xu. 2023. Foreground-aware task disentanglement and self-paced curriculum learning for domain adaptive object detection. *IEEE Transactions on Neural Networks and Learning Systems* (2023).
- [38] Yang Liu, Zhaoyang Xia, Mengyang Zhao, Donglai Wei, Yuzheng Wang, Siao Liu, Bobo Ju, Gaoyun Fang, Jing Liu, and Liang Song. 2023. Learning causality-inspired representation consistency for video anomaly detection. In *Proceedings of the 31st ACM international conference on multimedia*. 203–212.
- [39] Gengyu Lyu, Xiang Deng, Yanan Wu, and Songhe Feng. 2022. Beyond shared subspace: A view-specific fusion for multi-view multi-label learning. In *Proceedings of the AAAI conference on artificial intelligence*, Vol. 36. 7647–7654.
- [40] Gengyu Lyu, Weiqi Kang, Haobo Wang, Zheng Li, Zhen Yang, and Songhe Feng. 2024. Common-individual semantic fusion for multi-view multi-label learning. In *International Joint Conference on Artificial Intelligence*. 4715–4723.
- [41] Gengyu Lyu, Zhen Yang, Xiang Deng, and Songhe Feng. 2024. L-VSM: Label-Driven View-Specific Fusion for Multiview Multilabel Classification. *IEEE Transactions on Neural Networks and Learning Systems* (2024).
- [42] Erlin Pan and Zhao Kang. 2023. High-order multi-view clustering for generic data. *Information Fusion* 100 (2023), 101947.
- [43] Yalan Qin, Zhenjun Tang, Hanzhou Wu, and Guorui Feng. 2023. Flexible tensor learning for multi-view clustering with Markov chain. *IEEE Transactions on Knowledge and Data Engineering* 36, 4 (2023), 1552–1565.
- [44] Qiangqiang Shen, Tingting Xu, Yongsheng Liang, Yongyong Chen, and Zhenyu He. 2023. Robust tensor recovery for incomplete multi-view clustering. *IEEE Transactions on Multimedia* 26 (2023), 3856–3870.
- [45] Yinghui Sun, Xingfeng Li, Quansen Sun, Min-Ling Zhang, and Zhenwen Ren. 2024. Improved weighted tensor Schatten p-norm for fast multi-view graph clustering. In *Proceedings of the 32nd ACM International Conference on Multimedia*. 1427–1436.
- [46] Q Tan, G Yu, J Wang, C Domeniconi, and X Zhang. 2019. Individuality-and commonality-based multiview multilabel learning. *IEEE transactions on cybernetics* 51, 3 (2019), 1716–1727.
- [47] X Tan, C Zhao, C Liu, et al. 2024. A two-stage information extraction network for incomplete multi-view multi-label classification. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 38. 15249–15257.

- [48] A V Turukmane and R Devendiran. 2024. M-MultiSVM: An efficient feature selection assisted network intrusion detection system using machine learning. *Computers & Security* 137 (2024), 103587.
- [49] Kaixiang Wang, Xiaojian Ding, and Fan Yang. 2024. Non-Overlapped Multi-View Weak-Label Learning Guided by Multiple Correlations. In *Proceedings of the 32nd ACM International Conference on Multimedia*. 4690–4698.
- [50] Shuqin Wang, Yongyong Chen, Zhiping Lin, Yigang Cen, and Qi Cao. 2023. Robustness meets low-rankness: Unified entropy and tensor learning for multi-view subspace clustering. *IEEE Transactions on Circuits and Systems for Video Technology* 33, 11 (2023), 6302–6316.
- [51] Shuqin Wang, Zhiping Lin, Qi Cao, Yigang Cen, and Yongyong Chen. 2023. Bi-nuclear tensor Schatten-p norm minimization for multi-view subspace clustering. *IEEE Transactions on Image Processing* 32 (2023), 4059–4072.
- [52] X Wu, Q-G Chen, Y Hu, D Wang, X Chang, X Wang, and M-L Zhang. 2019. Multi-View Multi-Label Learning with View-Specific Information Extraction. In *IJCAI*. 3884–3890.
- [53] Wulin Xie, Xiaohuan Lu, Yadong Liu, Jiang Long, Bob Zhang, Shuping Zhao, and Jie Wen. 2024. Uncertainty-aware pseudo-labeling and dual graph driven network for incomplete multi-view multi-label classification. In *Proceedings of the 32nd ACM International Conference on Multimedia*. 6656–6665.
- [54] Jie Xu, Huayi Tang, Yazhou Ren, Liang Peng, Xiaofeng Zhu, and Lifang He. 2022. Multi-level feature learning for contrastive multi-view clustering. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 16051–16060.
- [55] Shuyu Yang, Yinan Zhou, Zhedong Zheng, Yaxiong Wang, Li Zhu, and Yujiao Wu. 2023. Towards unified text-based person retrieval: A large-scale multi-attribute and language search benchmark. In *Proceedings of the 31st ACM International Conference on Multimedia*. 4492–4501.
- [56] Chao Zhang, Huaxiong Li, Wei Lv, Zizheng Huang, Yang Gao, and Chunlin Chen. 2023. Enhanced tensor low-rank and sparse representation recovery for incomplete multi-view clustering. In *Proceedings of the AAAI conference on artificial intelligence*, Vol. 37. 11174–11182.
- [57] C Zhang, Z Yu, Q Hu, P Zhu, X Liu, and X Wang. 2018. Latent semantic aware multi-view multi-label classification. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 32.
- [58] Fangwen Zhang, Xiuyi Jia, and Weiwei Li. 2020. Tensor-based multi-view label enhancement for multi-label learning. In *IJCAI*. 2369–2375.
- [59] Jia Zhang, Zhiming Luo, Candong Li, Changen Zhou, and Shaozi Li. 2019. Manifold regularized discriminative feature selection for multi-label learning. *Pattern Recognition* 95 (2019), 136–150.
- [60] Min-Ling Zhang and Zhi-Hua Zhou. 2013. A review on multi-label learning algorithms. *IEEE transactions on knowledge and data engineering* 26, 8 (2013), 1819–1837.
- [61] Yuhan Zhang, Bo Wu, Wen Li, Lixin Duan, and Chuang Gan. 2021. STST: Spatial-temporal specialized transformer for skeleton-based action recognition. In *Proceedings of the 29th ACM International Conference on Multimedia*. 3229–3237.
- [62] Y Zhang, J Wu, Z Cai, et al. 2020. Multi-view multi-label learning with sparse feature selection for image annotation. *IEEE Transactions on Multimedia* 22, 11 (2020), 2844–2857.
- [63] Dawei Zhao, Qingwei Gao, Yixiang Lu, and Dong Sun. 2022. Learning view-specific labels and label-feature dependence maximization for multi-view multi-label classification. *Applied Soft Computing* 124 (2022), 109071.
- [64] Dawei Zhao, Qingwei Gao, Yixiang Lu, and Dong Sun. 2022. Non-aligned multi-view multi-label classification via learning view-specific labels. *IEEE Transactions on Multimedia* 25 (2022), 7235–7247.
- [65] Qiyu Zhong, Gengyu Lyu, and Zhen Yang. 2024. Align while fusion: A generalized nonaligned multiview multilabel classification method. *IEEE Transactions on Neural Networks and Learning Systems* (2024).
- [66] Aichun Zhu, Zijie Wang, Yifeng Li, Xili Wan, Jing Jin, Tian Wang, Fangqiang Hu, and Gang Hua. 2021. Dssl: Deep surroundings-person separation learning for text-based person retrieval. In *Proceedings of the 29th ACM international conference on multimedia*. 209–217.
- [67] Changming Zhu, Duoqian Miao, Zhe Wang, Rigui Zhou, Lai Wei, and Xiafen Zhang. 2020. Global and local multi-view multi-label learning. *Neurocomputing* 371 (2020), 67–77.
- [68] Xiaofeng Zhu, Xuelong Li, and Shichao Zhang. 2015. Block-row sparse multiview multilabel learning for image classification. *IEEE transactions on cybernetics* 46, 2 (2015), 450–461.